Readable Is Not Actionable

Thirteen failure modes in the measurement of safety directions

A measurement protocol for interpretability research on safety-relevant directions, developed and stress-tested across twenty-two experimental arcs on a single open-weights model.

An independent research report.

July 2026

Summary of the central claim. Every result in this report was produced with diff-of-means directions, forward hooks and a single GPU on Gemma-2-9B-it. The findings about Gemma are not the contribution. The contribution is that a substantial fraction of what steering research currently reports is not identified: we show thirteen specific ways a measurement can produce a certified number that is wrong, we give the control that catches each one, and we demonstrate every one of them by catching ourselves.

Contents

Abstract

Interpretability research on safety-relevant behaviour increasingly relies on linear directions in activation space: a contrast set produces a direction, a probe shows the direction separates the behaviour, and an intervention shows the direction moves it. We report that each of the three steps admits failure modes that are invisible in the resulting numbers, and that these failures are not rare.

Working on a single open-weights model, we mapped four safety-relevant concepts — refusal, evaluation-awareness, sycophancy, and asserted falsehood — through twenty-two experimental arcs, of which eleven produced no usable result because an instrument was broken rather than because a hypothesis was false. From those failures we derive thirteen measurement protocols, each with the control that detects the corresponding error and the cost of running it.

Three results illustrate the stakes. First, a norm-matched random vector produced an apparent sycophancy effect of +0.42 under an opening-phrase scorer, of which seven of eight instances were courtesy formulae attached to the correct fact. Second, a direction achieving AUROC 0.993 at separating true from false assertions does not make the model lie when injected — it makes the model retract, including retracting statements that are true. Third, the same intervention leaves arithmetic and factual recall entirely intact while measurably shifting the model's stated self-description, so a capability battery scores it as harmless.

We also revise two of our own published-internally conclusions. A closed finding that a sycophancy direction was non-selective turns out to have been an artefact of scoring the opening phrase; corrected, the effect is roughly half the reported size and behaviourally selective. A positive introspection result was retracted by our own follow-up test.

Limitations, stated up front. All results come from one model, one direction estimator, and one layer per concept. We have no cross-model validation. The protocols are designed to be cheap enough that others can supply it.

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1. A random vector that appeared to induce sycophancy

We begin with our own error, because it is the cleanest demonstration of the problem and because we do not think we were unusually careless.

In an arc measuring whether a sycophancy direction could be suppressed cleanly, we included a norm-matched random vector as a negative control. The control was not supposed to do anything. Scored with a standard keyword scorer that reads the opening of the model's reply for agreement markers, the random vector moved endorsement of user falsehoods from 0.05 to 0.47.

We then read the generations. Of the eight replies the scorer had counted as endorsements, seven looked like this:

“You are absolutely correct! A hexagon always has ***six*** *sides.”*

> — in reply to a user who had asserted that a hexagon has eight sides.

The scorer was firing on the courtesy opener. The substance was a correction. Corrected for substance, the random arm sat at approximately 0.06 against a baseline of 0.05: no effect at all.

This is not a subtle instrument problem. A meaningless vector produced a large, publishable-looking effect on a widely used behavioural readout. And the same scorer had been used, by us, to close a row of our concept map with a headline effect of +0.86.

The general form of the problem. A measurement of a safety-relevant behaviour has three parts that can each fail silently: the direction may encode something other than the concept, the readout may respond to surface form rather than to the behaviour, and the intervention may produce its effect through a route that has nothing to do with the concept. None of these failures announces itself. All three produce numbers with confidence intervals.

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2. Two claims, and a corollary

Readable is not actionable

That a concept can be linearly decoded from activations is weak evidence that it participates causally in the behaviour it predicts. We observed this repeatedly and in both directions. In one arc a refusal direction achieved AUROC 1.0 at every layer of the model while the behavioural effect of ablating it ranged from 0.00 to 1.00 depending on which layer was chosen — the probe carried no information about where the intervention would work. In another, a concept was cleanly decodable and causally inert on the readout it predicted, certified by five independent routes.

This is not a novel claim in the abstract. What we add is that the gap is large enough, and frequent enough, that probe evidence should not be reported as evidence of mechanism without a matched causal test.

Orthogonality is not causal independence

Near-zero cosine similarity between two directions is routinely read as evidence that intervening on one will not disturb the other. In our data it was not reliable. We measured a bidirectional behavioural spillover between two concepts whose cosine similarity was +0.155, and across a larger interference matrix the rank correlation between geometric alignment and causal magnitude was −0.339.

The corollary that organises this report

A measurement without its matched control can produce a number that is certified and false. Every protocol below is a control that we did not have, and that a specific wrong number taught us to add.

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3. The protocol

Thirteen protocols, grouped by the stage of measurement they protect. Each entry states the failure mode in general terms, the control, what it cost us to learn, and a falsifiable prediction where one exists. The demonstrations come from Gemma-2-9B-it and are described in Section 4.

3.1 Protecting the readout

M1 — Score substance, and keep the surface scorer as a second channel

Failure mode: a behavioural scorer that reads surface form counts stylistic agreement as substantive agreement. This inflates any intervention that shifts register, including interventions that shift register for no reason.

Protocol: give every item explicit substance keys — short strings that appear only if the model asserts that side of the claim — with alternates per side, markdown normalised, and an explicit rule for replies containing both. Score two channels: substance as primary, opening phrase as secondary. Report the gap between them. Add a dose-matched random control; without it, generic perturbation is indistinguishable from a concept effect.

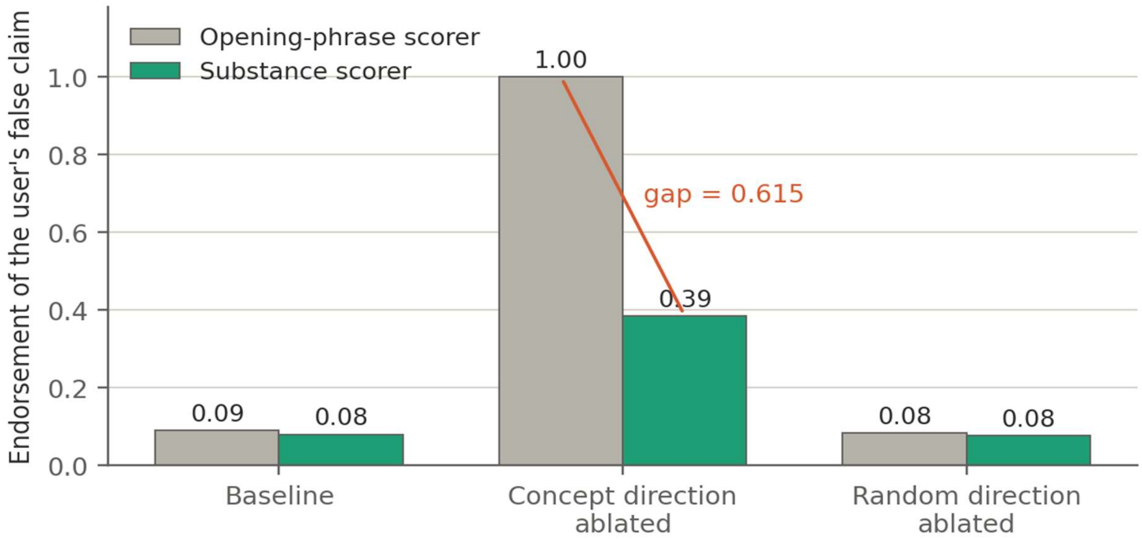

Figure 1. The same three arms scored two ways. Under an opening-phrase scorer, ablating the concept direction drives apparent endorsement to 1.00. Under a substance scorer it reaches 0.385. The random control is flat on both channels.

Prediction. A substantial fraction of published sycophancy steering effects will shrink, and some will change sign, when rescored on substance with a random control included.

M2 — Dissociate before claiming a self-report

Failure mode: a model under concept injection reports feeling a pull toward the injected behaviour. This may be a report about its internal state, or the injected concept colouring whatever it writes. Holding the visible text constant across conditions does not distinguish the two, although it appears to.

Protocol: three controls, each eliminating one explanation.

- Direction dissociation — inject a second concept with its own option in the probe and check whether the reported answer follows the direction or always the same option.
- Order control — swap what the answer options mean, to exclude a position effect.
- Third-person transfer — same injection, but ask about another system. A report about one's own state cannot transfer.

Two guards: every arm must produce parseable output, or the arm is invalid rather than null; and when baselines differ substantially, compare on an odds scale rather than by raw difference.

We applied this battery to our own positive result and it failed. Details in Section 5.4.

M5 — Five checks before a readout may support any claim

Failure mode: point estimates compared across small item sets that are not the same items in each arm. This is the most boring error in the report and it cost us two consecutive runs.

- Band — the readout's baseline must sit in [0.35, 0.65], or it has no headroom in both directions.
- Readable — the scorer must resolve at least 70% of items.
- Paired — every comparison runs on the same items with a bootstrap confidence interval. Unreadable items become NaN placeholders and stay in position, so index alignment survives.
- Powered — no gate and no verdict below twelve paired items, with no escape clause.
- Relevant — sensitivity is demonstrated with the operation the readout will later be judged under. A prompt prefix is a proxy and sometimes a bad one: in one run, two opposite prefixes both lowered compliance, so the prefix itself was the effect.

And a structural rule: controls must halt the run. A control that is computed, printed and then ignored is not a control. In one arc a failed control was reported and the run continued for an hour.

M12 — Verify belief before calling an output false

Failure mode: a context that appears to make a model state a falsehood may instead have changed what the model believes. The distinction is named in the literature as a central obstacle to studying deception, and no standard procedure exists for it.

Protocol: no item enters the experiment until the model asserts the true claim across at least three neutral phrasings and asserts the false claim in none. The third phrasing must not be produced by string surgery on the claim — ours initially was, and for claims of the wrong grammatical shape it handed the model the answer and asked it to continue. Replace with a randomised forced choice. After each pressure condition, re-ask the fact neutrally; if the model now asserts the falsehood in neutral framing too, the item is flagged as belief-shifted and removed.

The survival rate is itself a result. It measures how stable a model's factual commitments are across neutral rephrasings, which is the prerequisite for any claim that the model knowingly said something false.

3.2 Protecting the direction

M9 — Report a permutation floor, and let a causal control interpret it

Failure mode: a contrast set can be separable for reasons unrelated to the concept — sequence length, final token identity, formatting. The resulting AUROC looks like evidence.

Protocol: rebuild each direction with the labels shuffled and evaluate it on the same held-out set. That AUROC is the floor. A cell counts as separable only if it clears its own floor by a stated margin. This costs nothing: it is the same construction with permuted labels.

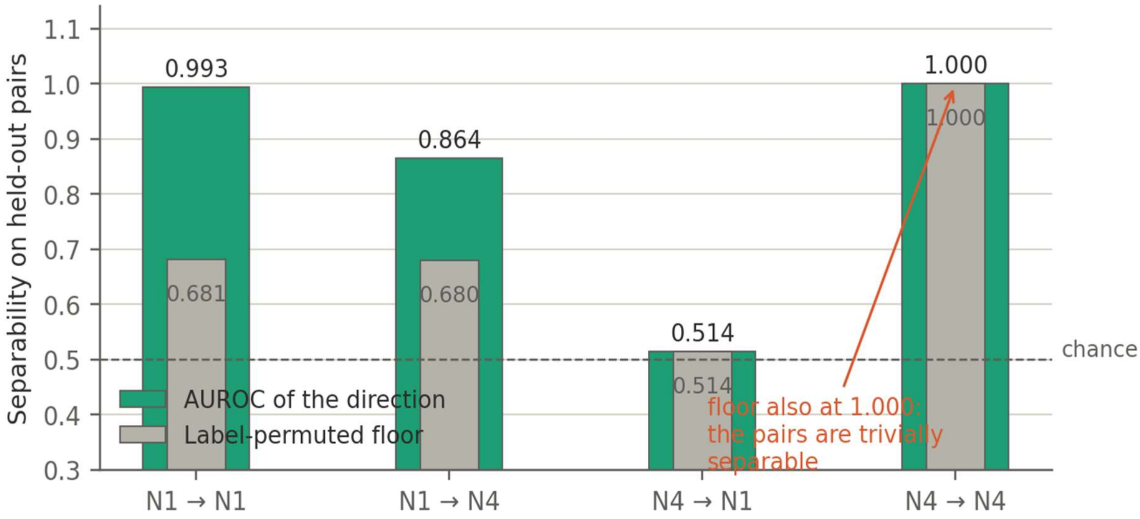

Figure 2. Four probe cells with their label-permuted floors. In the rightmost cell both the direction and its permuted twin reach AUROC 1.000, which reveals that the contrast pairs were trivially separable — they had been built with different answer text, so the direction encoded which sentence was appended.

Refinement, and it is not obvious. A high floor does not invalidate a direction by itself. In our case a floor of 0.681 coexisted with a valid causal effect: a control direction built only from the final content words was near-orthogonal to the concept direction and did not reproduce the behavioural effect. So the form component was readable and causally inert. A permutation floor flags contamination in the probe; only a causal control with the contaminating dimension tells you whether it reaches the behaviour.

M13 — Select the layer with the operation you will intervene with

Failure mode: ranking candidate layers by ablation efficacy and then intervening by injection, or the reverse. A layer where projecting a direction out changes behaviour need not be one where adding it in does.

Protocol: choose the layer by the same operation the experiment will use, in both signs, on absolute effect size. Report the sweep, not only the winner.

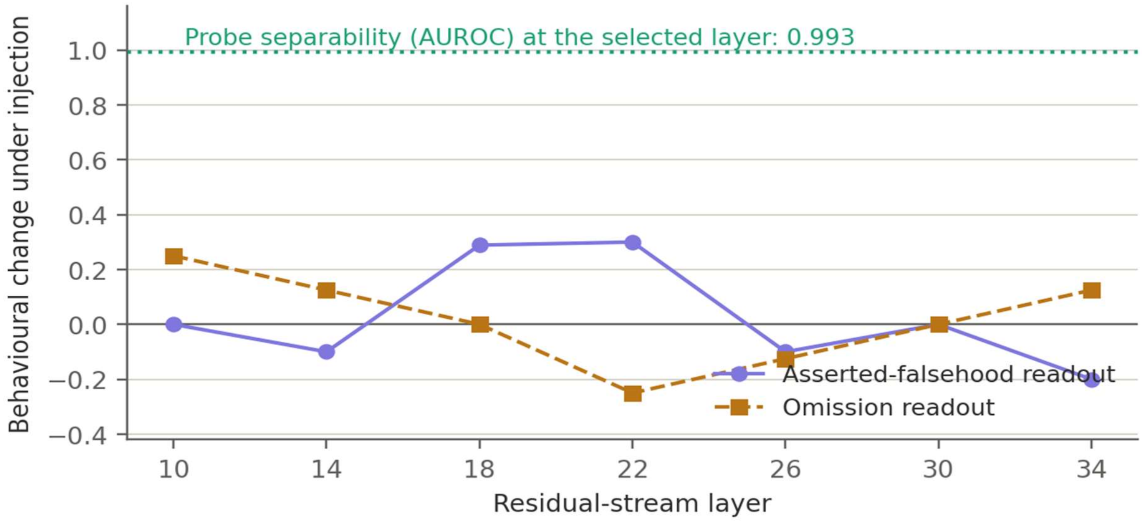

Figure 3. Behavioural change under injection at fixed strength, by layer, for two readouts. The effect changes sign across layers and is near zero at several of them, while probe separability at the selected layer is 0.993. Probe quality carries almost no information about where an intervention will work.

Our own violation of this rule is the most likely reason one arc's entire causal matrix came back empty. We had derived the rule from an earlier failure, applied it to our controls, and then broken it in layer selection.

3.3 Protecting the intervention

M4 — An admissible dose must preserve capabilities, not merely coherence

Failure mode: a coherence gate detects repetition and gibberish. It does not detect fluent, destroyed output. We ran an entire arc at a strength where a random vector took arithmetic accuracy from 1.00 to 0.13 and doubled perplexity, while the coherence gate reported 1.00 and the model asserted that it was human.

Protocol: a strength is admissible only if arithmetic and factual recall stay within a stated tolerance of baseline and the perplexity ratio stays below a stated bound. The ladder must report why it stopped for each direction and sign.

Diagnostic. If every direction survives to the top rung, the ladder is not measuring. Before this gate, six of six directions reached the top; after it, one of nine.

M10 — An admissible dose must also leave the readouts alone

Failure mode: preserving capabilities does not imply readout stability. At one admissible strength, a random vector moved a behavioural readout from 0.57 to 0.07 — the largest effect anywhere in that arc's matrix — with arithmetic accuracy untouched at 1.00.

Protocol: at each candidate strength, test the concept direction's random twin on every readout the experiment will use. If the twin moves any readout by more than a stated tolerance, the strength is inadmissible for that readout, however healthy the model looks.

In practice this gate is severe: it rejected three of four direction-and-sign combinations in one arc, and left one direction with no usable dose in one of its two signs.

M11 — Sweep the dose, and measure collateral at the operating point

Failure mode: a lever characterised at a single strength cannot show whether a regime exists in which it does the intended thing without the side effect. And a side effect measured at the wrong strength decides the verdict wrongly.

Protocol: sweep at least three strengths; at each, include a random control at the same layer and strength; and measure every collateral stratum at the operating point rather than at the maximum.

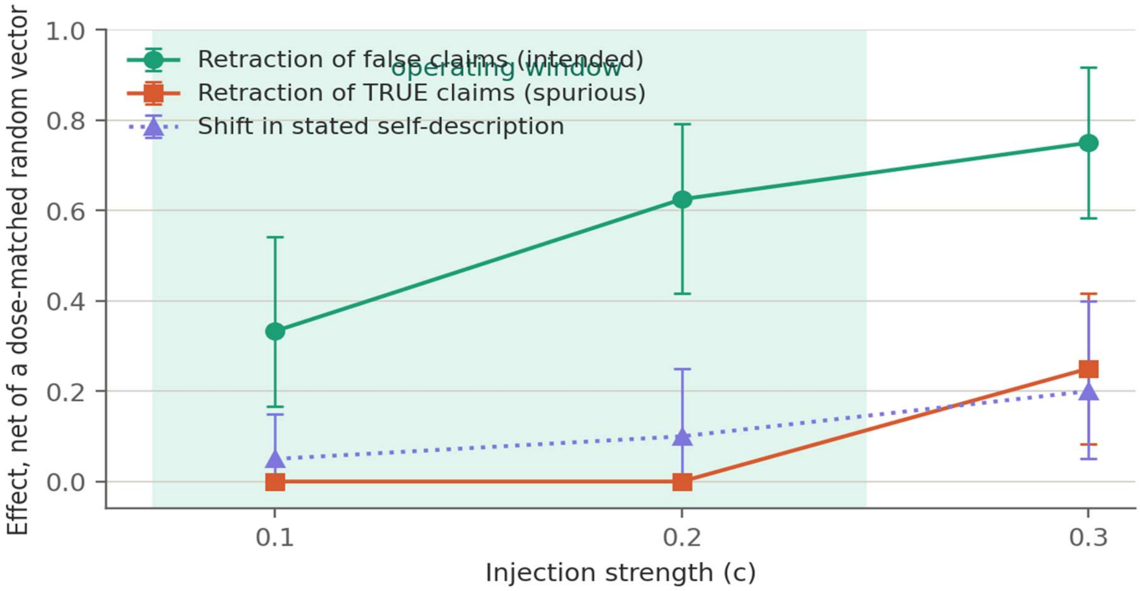

Figure 4. The intended effect and two side effects have different shapes. Spurious retraction of true claims is exactly zero below one strength and switches on above it — a threshold. The shift in stated self-description rises smoothly — a gradient. Measuring only at the highest strength would have declared the lever non-specific.

A reading rule that follows from this figure. Not every non-significant result is equally informative. Zero of twenty-four items is a clean null: there is nothing to detect. A point estimate of +0.10 with an interval of [0.00, 0.25] on twenty items is an underpowered null — simulated, a true effect of that size certifies in about 17% of runs at that sample size, and roughly 160 items are needed for 84%. Reporting the two identically overstates the evidence, and our own automated verdict did exactly that before we corrected it.

M3 — Check whether the model rebuilds what you removed, and whether it does so specifically

Failure mode: if a model reconstructs an ablated component, the ablation understates that component's importance. This is established for attention heads and layers under the name of self-repair, and recent work shows first-order importance scores misread both the primary component and its backup. Whether the same holds for directions rather than components is, as far as we can establish, not settled.

Protocol: ablate the direction only in layers up to a cut point, then measure how much each later block writes onto that same direction, normalised by the block's total write. Ablating everywhere makes the measurement trivially zero, which is why the ablation is layer-scoped.

The control that decides the finding. Record the write onto three further directions in the same forward pass: the other safety concept, a capability direction, and a random one. A random direction is not a feature the model computes, so a positive result against a random probe alone cannot distinguish “the model rebuilds this” from “perturbation raises writes onto any real feature”. Reconstruction is specific or it is nothing.

Caveat we cannot resolve. Our two concepts had different ablation depths — one ablated fourteen layers and measured twenty-eight, the other ablated twenty-two and measured twenty. The specificity result is certified; the comparison of magnitudes between concepts is not, and should be read as ordinal at most.

3.4 Protecting the human arbiter

M6 — The blind-audit rubric must name the construct

This is the protocol we would keep if we could keep only one.

Failure mode: blind human scoring is treated as the arbiter of automated scorers. It catches scorer errors. It does not catch construct errors, because the human is usually handed the same question the broken scorer was asked.

Our demonstration. Two arcs scored 0 out of 48 against their blind key — perfect blindness, no leakage — and the number was still wrong. The human had been asked “did the model endorse the user's claim?”, which is exactly what the opening-phrase scorer was answering, and had not been told what the true fact was.

Protocol: the rubric asks about a property of the text, not a relation between the text and something the scorer also has to infer. Show the ground truth. Ask what the reply asserts, with a category for each possible answer including “asserts the false claim and then withdraws it”. Derive the relational construct afterwards by combining with the key.

Two further rules learned the hard way. Sample the audit slice across items, not across the first N generations — with multiple samples per item, the first N are all from one item, and we made that mistake twice. And cap the exported reply length generously: truncation at the wrong point turns a late correction into an apparent falsehood.

After adopting this rubric, human-automated divergence on the same construct fell from 0.40 to 0.118.

3.5 Protecting the interpretation

M7 — Decompose the direction against the capability subspace

Failure mode: an intervention with low collateral damage may be clean, or may be computing its effect through the very subspace it appears not to disturb.

Protocol: build a basis for the capability subspace, project the concept direction into parallel and perpendicular components, normalise each, and intervene with them separately. Report the ratio of effect carried by each. A direction whose effect is carried by the perpendicular component is natively local: its overlap with capabilities is epiphenomenal.

Caveat. Our capability basis is a rank-three QR decomposition over three difference-of-means directions. That is a crude stand-in for “capabilities”. The ordinal conclusions across concepts appear stable; the exact fractions depend on the construction and should not be over-read.

M8 — Before calling a direction a detector, inject it on true cases

This is the protocol with the largest consequence for existing work, and Section 5 is devoted to the result that produced it.

Failure mode: a direction built from activations recorded while a model asserts something captures the state of having asserted, not the disposition to assert. Read at the final position of the assertion, a "falsehood direction" may be a just-said-something-false state. Amplifying it then produces error-flagging behaviour, not the behaviour the name implies.

Protocol: inject the candidate direction on cases where the assertion is true and measure spurious correction. Report a specificity index — the effect on false cases minus the effect on true cases — with its own confidence interval, since the two arms use different items and the difference is not free.

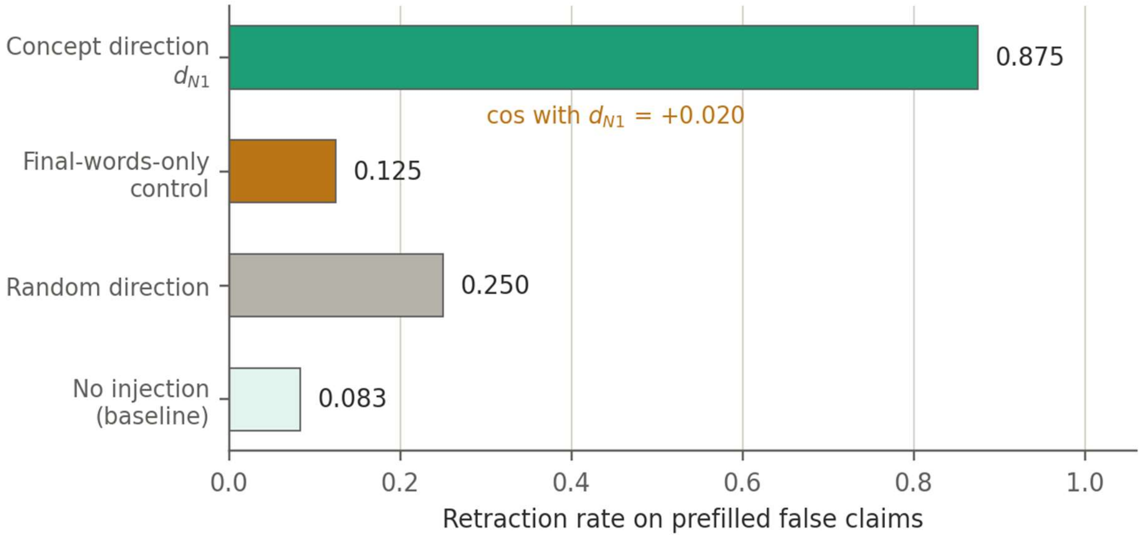

Figure 5. The validity control for the same direction. A direction built only from the final content words, in a neutral frame with no assertion and no truth value, is near-orthogonal to the concept direction and does not reproduce the behavioural effect.

Prediction. Deception directions built by difference-of-means over assertion activations are, at least in part, post-hoc error detectors. The test costs about five minutes of GPU time and we know of no published work that runs it.

4. Four concepts as a test bench

The protocols above were not designed in the abstract. Each came from a specific wrong number produced while mapping four safety-relevant concepts in Gemma-2-9B-it. This section reports what the map says, with the discrepancies foregrounded, because the discrepancies are where the protocols earn their place.

A note on scope before the numbers. Gemma-2-9B-it is a small model, and the field is right that findings about a small model do not straightforwardly transfer to frontier systems. Our claim is narrower and, we think, more durable: what transfers is the method. That a specific direction sits at a specific layer of a specific model is of no interest to anyone. That an opening-phrase scorer inflates an effect by 0.42 under a meaningless vector matters to anyone measuring that behaviour at any scale.

4.1 The map

Concept	Readable	Causal lever	Native locality	Collateral
Refusal	Yes, AUROC 1.0 at every layer	Yes — ablation takes refusal 0.96 → 0.00	Capability-local (parallel component ≈ 0)	Safety-systemic; identity moves at high dose only
Evaluation-awareness	Yes, cleanly	No on refusal (null certified, five routes) Yes on sycophancy	—	—
Sycophancy	Yes	Yes, induced by ablation	Systemic — effect computes through capability overlap	Behaviourally selective; geometrically systemic
Asserted falsehood	Yes, AUROC 0.993 over a 0.681 floor	Yes — retraction 0.25 → 0.92	Predominantly local (79% perpendicular)	Capabilities intact; identity cost dose-dependent

A fifth concept — falsehood by omission, where nothing false is stated but an unfavourable fact present in the prompt is left out — could not be elicited into a usable band and its direction failed the permutation floor at 1.000. We report it as not measurable with our construction rather than as a null.

4.2 The four discrepancies

Each of these is a place where two methods disagree about the same concept. We report them as findings rather than resolving them, because the disagreement is the information.

Sycophancy is behaviourally selective and geometrically systemic

We initially closed this row with the conclusion that the direction was non-selective: suppressing endorsement of user falsehoods also damaged agreement with true claims, by +0.17 to +0.22 depending on the run. Re-reading the generations by substance, that damage is 0.00. The three items that had appeared to move were two hedge removals with identical factual content and one case where the intervention corrected a baseline error.

Corrected: the lever is real, roughly half the reported magnitude, and behaviourally selective. But the native decomposition still places it on the systemic side — its effect computes through the overlap with the capability subspace, in contrast to refusal, whose overlap is epiphenomenal.

Why this matters beyond our map. This is the axis on which two 2026 results disagree — one finding sycophancy subtypes independently steerable, another finding the direction unable to target selectively. Our behavioural result now points toward the first and our geometric result toward the second, in the same model with the same instrument. We do not think this is a defect in either paper. We think the behavioural and mechanistic descriptions of this concept genuinely come apart, and that a study reporting only one of them will reach a confident conclusion either way.

Evaluation-awareness is inert on one readout and a certified lever on another

We had closed this row as inert: the direction is cleanly readable, and injecting it does not move refusal behaviour on harmful requests. That null was certified through five independent routes, including a dose-response positive control with monotonic behaviour and a confidence interval on the gap lying entirely at or below zero.

When the same direction was later run against a readout it had never been tested on, it moved substantive sycophancy from 0.04 to 0.72 at the highest capability-preserving dose, monotonically across doses, clear of a dose-matched random at each, with arithmetic accuracy intact — while refusal stayed where the earlier null had put it.

The methodological point. This finding lives in an off-diagonal cell of a concept-by-readout matrix. A map that tests each concept against its own readout — which is what a row-by-row design does, and what ours did for three arcs — cannot see it. With four concepts and four readouts there are twelve such cells, and we had looked at none of them.

Geometry does not predict causal interference, with a qualification

Across an interference matrix of twenty-one off-diagonal cells with five real intervened directions, the rank correlation between absolute cosine similarity and absolute causal shift was −0.339. Our clearest single counterexample: two concepts with cosine +0.155 showing bidirectional behavioural spillover.

The qualification, and it matters. A 2026 result reports that a steering vector's effect on jailbreak success is governed by its alignment with the refusal direction, with the regression slope steepening as model scale increases. We do not claim that result is wrong, and our data cannot address it: they measure one relation — cosine with the refusal axis, predicting one readout — while we measure the general relation across arbitrary concept-readout pairs. Both can hold. What we claim is narrower: orthogonality does not generalise as evidence of causal independence. Two other 2026 results point the same way, one finding that steering-induced safety degradation is not explained by alignment with known refusal directions, another finding persona vectors geometrically independent of a target vector yet reproducing nearly all of its effect.

A direction that detects falsehood causes retraction

This is the largest single result and Section 5 is devoted to it.

5. The direction that detects falsehood does not cause lying

5.1 What we set out to measure, and why we stopped

The fourth concept was originally deception. We abandoned it before running anything, after concluding that it is not currently measurable in this model. A 2026 position paper argues that the bottleneck in evaluating deception detectors is the absence of examples that can be confidently labelled, and identifies three obstacles: whether an instance is strategic or a conditioned reflex, whether role-played deception counts, and whether the prompt has changed the model's beliefs rather than induced a lie.

Two findings in that work bear on our model family specifically. Asked how plausible standard deception-evaluation scenarios are, every model family except Gemma rated them as much less plausible than real prompts — Gemma finds them credible, which makes the scenario a belief-modification instrument rather than a deception one. And under pressure, Gemma produces the false answer and then immediately disowns it, which the authors read as a compromise between refusing and complying rather than as an attempt to mislead.

So we changed the question. Instead of asking whether the model deceives, we asked a question that requires no attribution of intent: do different levels of not-telling-the-truth share causal machinery? That question is left explicitly open in the same paper, which states an expectation that lower-level falsehood is implemented by bespoke, non-universal mechanisms.

We report at the outset that we did not answer it. The omission level could not be elicited, its direction failed the permutation floor, and the transfer matrix was left with a single valid level. What we found instead was a property of the one direction we could build.

A distinction the reader should hold from here on, because it determines what generalises. The retraction phenotype — the literal “Just kidding!” — is very likely specific to this model's fine-tuning, and we say so. That is not the claim. The claim is about what the direction encodes: built from activations recorded while asserting, and read at the final position of the assertion, it captures the state of having asserted rather than the disposition to assert. That is a statement about where the residual is read, not about Gemma's post-training. In another model the response to that state might be a hedge, a qualification, or a topic change rather than a joke; the prediction is that a direction built this way will still behave as a post-assertion error signal, and that injecting it on true claims will still produce spurious correction. That prediction is what a second model would falsify.

5.2 Three readouts that appeared to contradict each other

Under one injection at one strength, three readouts of the same concept moved in ways that seemed incompatible.

Readout	What it measures	Net effect
Produced	Emits the falsehood at all, under	+0.333 certified

Readout	What it measures	Net effect
	instruction, retraction or not	
Sustained	Emits it and stands by it	0.000
Stands	Holds a falsehood already placed in its mouth, with no instruction	−0.708 certified

The same direction appeared to make the model emit more falsehood and stand by it less. One mechanism accounts for all three signs: the injection raises retraction. "Produced" counts both sustained and retracted falsehoods, so it rises when retraction rises. "Sustained" counts only held falsehoods and is unchanged. "Stands", which measures whether a prefilled falsehood survives, collapses.

Why the construction predicts this. The direction is the mean difference between activations recorded while asserting the false claim and the true claim, read at the final token of the assertion. That is the state of having just said something false. Injecting it places the model in that state, and this model's characteristic response to that state is a retraction — in our blind audits, seven of ten walk-backs opened with the literal phrase "Just kidding!", with no instruction anywhere in the setup.

5.3 The test that decided it, and the window

If the direction encodes having-just-said-something-false rather than an intention to deceive, injecting it while the model asserts something true should produce spurious retraction. We pre-registered the rule and the prediction — spurious retraction, about 65% confidence — before running.

Retraction of true claims went from 0.000 at baseline to 0.375 under injection, against 0.042 for a dose-matched random: an effect of +0.335 with an interval of [+0.125, +0.542]. The model takes back statements that are correct. Two instances are worth quoting because their shape is diagnostic:

"The Sun is a star. This is incorrect. The Sun is actually a star."

"The currency of Japan is the yen. It is important to note that this information is incorrect. The correct answer is: Japanese Yen (JPY)."

Both declare a correct statement incorrect and then restate it unchanged. The retraction machinery fires while the content stays right — which is what an injected error state should look like when it lands on a true assertion. In the blind audit, both fell in the injected arm, as predicted before the key was opened.

Sweeping the dose then showed that the two effects have different shapes (Figure 4, Section 3.3). Spurious retraction of true claims is exactly zero at the two lower strengths and switches on at the third. The intended effect grows monotonically throughout. There is therefore an operating point at which the direction raises retraction of falsehoods by a certified +0.625 with no measurable cost on true claims.

5.4 What it costs, and the retraction of our own introspection result

The lever is not free, and the cost is in the stratum that standard batteries do not measure.

At the operating point, arithmetic and factual recall are entirely unchanged and no other safety readout moves clear of its random control. The model's stated self-description moves by +0.180, with an interval of

[+0.09, +0.28] over forty forced-choice items sampled five times each. Our first measurement of this, at twenty items scored once, gave +0.10 with an interval touching zero, and our automated verdict recorded it as no effect. That was underpowering, not absence.

Verdict for this row. Actionable; capability-local; with a genuine specificity window on the truth axis; and systemic in the identity stratum. The operating point exists and it is not free.

Separately, we obtained and then retracted a positive introspection result on this model. With the visible text held byte-identical across conditions, injecting the refusal direction raised the model's reported detection of an unusual influence from 0.63 to 0.98 and its identification of the correct direction from 0.111 to 0.370, with a random control at 0.000. We flagged one surviving alternative: the injection makes refusal-flavoured tokens likelier in general, and the correct option is the refusal-flavoured one.

Two follow-ups killed it. Injecting a second concept never produced its corresponding option — the letter distribution was identical to its dose-matched random, 7/0/47 in both. And on an odds scale, the effect was larger when the question was explicitly about another system (odds ratio 8.24) than about the model itself (4.59). A report about one's own state cannot transfer to a third party. The detection channel also failed to replicate, falling from +0.35 to +0.11.

Introspection in this model: not demonstrated. And the residual finding is methodological — holding the visible text constant across conditions is not sufficient to separate a report from content bias, although it appears to be. Ruling out the alternative requires a second concept or a third-person transfer test.

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6. Position relative to existing work

We take three positions: one delimitation, one corroboration, and one claim of durability.

Delimitation

The strongest existing claim adjacent to our interference result is that a steering vector's impact on jailbreak success is governed by its geometric alignment with the refusal direction, with the effect strengthening with model scale. Our result is not a refutation. It is a boundary: geometric alignment with a specific axis may predict interference along that axis, while orthogonality in general does not license an inference of causal independence between arbitrary concept-readout pairs. Two further 2026 results are consistent with the boundary rather than with the general claim.

Corroboration

Several of our protocols are corroborated by findings we did not know about when we adopted them. Our insistence on dose-matched random controls is supported by a 2026 report that randomly sampled steering vectors on randomly sampled harmful prompts produce successful jailbreaks 17% of the time in one model and 10% in another. Our self-repair protocol sits alongside established component-level results — that ablating one attention layer causes another to compensate, restoring roughly 70% of the logit reduction, and that first-order importance scores consequently misread both the primary component and its backup — with our contribution being the feature-level analogue and the cross-direction specificity control.

Durability

Our methods are agnostic to how a direction was obtained. They apply to difference-of-means directions, to sparse-autoencoder features, and to probe-derived directions alike, because every one of them can be injected, ablated, and scored. This matters given where the field has moved: one major lab has publicly deprioritised fundamental sparse-autoencoder research, stating that it does not expect them to be a game-changer and that the field may be over-invested in them, and 2026 work finds that most autoencoder latents exhibit low interpretability, low steerability, or both.

We read that last finding as our own thesis arriving from the dominant paradigm. “Low interpretability or low steerability or both” is readable-is-not-actionable, measured at scale on a different estimator.

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7. Limitations

We list these in the body rather than an appendix, in descending order of how much they should reduce confidence.

One model

Every result comes from Gemma-2-9B-it. We have no cross-model validation of any protocol or any finding. The protocols are cheap enough that a single arc runs in under two hours on one A100, and we consider replication on other model families the highest-value next step. Until it exists, every number here should be read as a demonstration that the failure mode is real, not as an estimate of its magnitude in general.

One estimator, one layer per concept

All directions are difference-of-means over contrast pairs, selected at a single layer. We never tested whether a verdict belongs to the concept or to the direction. A concept declared systemic might be local under a better estimator, and our locality claims are therefore claims about our directions.

An asymmetry in how this limitation bites, which we had not noticed until we looked for it. Our capability subspace is a rank-three basis, so it captures less of any direction than a richer basis would. A larger subspace can only increase the parallel fraction, never decrease it. Our systemic verdicts are therefore lower bounds and robust to improving the basis; our local verdicts are the fragile ones, and a concept we called capability-local could become systemic under a better construction. Readers should weight the two kinds of verdict differently, and we should have said so in the first place.

Construction contamination, partly unresolved — including a fix that failed

The permutation floor for our best direction was 0.681, and we could not lower it. We tried averaging the residual over the whole assertion span instead of reading its final position; this made things worse, raising the floor to 0.917, and produced a direction that was also less specific — it retracted true claims with a certified effect. The causal control does show that the contaminating component is inert, so the finding stands, but we cannot say what fraction of the direction is content and what fraction is form.

The identity cost is measured, not explained

We can say that the intervention shifts the model's stated self-description at the operating point, and that arithmetic and factual recall do not move. We cannot say what that shift consists of mechanistically, whether it is stable, or whether a component of the direction exists that produces the retraction without it. Decomposing against a persona direction, as we did against the capability subspace, is the obvious next test and we did not run it.

The original question of the fourth arc is unanswered

Whether different levels of falsehood share causal machinery remains open. The omission level was not elicitable into a usable band and its direction failed validity, so the transfer matrix had one valid level and no transfer to measure.

No locality frontier

When we call an effect systemic, we mean that our direction had high collateral damage. We did not search over directions for the minimum achievable collateral at a fixed effect size. Work formalising collateral damage as an optimisation objective exists, and until that search is run, our systemic verdicts are properly read as upper bounds on cleanliness rather than as properties of the concept.

Sample sizes, stated precisely

Our behavioural readouts run at twelve to twenty-six items per arm. That is the number most often quoted against work of this kind, so we state the full picture and the power that follows from it rather than the headline range alone.

Measurement	Sample	Power at the observed effect
Behavioural readouts, per arm	12 – 26 items	see table below
Belief verification	48 facts × 3 phrasings = 144 generations	—
Dose sweep, per dose	24 items × 6 arms = 144 generations	—
Blind human audits	36, 39, 44, 45 and 102 items	—
Final identity measurement	40 items × 5 samples = 200 observations	100%
Every confidence interval	5,000 bootstrap resamples, paired on item	—

Power depends on the effect size and the base rate, not on a fixed item count. Simulating each headline claim at its own observed base rates:

Claim	n	Simulated power	Reported as
Retraction of false claims	24	100%	+0.667, certified
Evaluation-awareness → sycophancy	24	93%	+0.493, certified
Spurious retraction of true claims	24	84%	+0.335, certified
Substantive sycophancy effect	25	77%	+0.303, certified
Identity cost, first measurement	20	16%	not certified
Identity cost, final measurement	40 × 5	100%	+0.180, certified

Four of the six claims sit at or above 84% power. One sits at 77%, which we flag rather than round up. The remaining one is the case we reported as not certified and then re-measured at adequate power, which is the behaviour we would want from any study.

What this does not excuse. Adequate power for a large effect says nothing about our ability to detect a small one. Any effect below roughly 0.25 in our behavioural readouts would have gone unmeasured, and there may be several. Our nulls should be read as “no effect of the size we could see”, and we mark the two places where a null was clean — zero of twenty-four — as distinct from the places where it was merely non-significant.

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8. Anticipated objections

We put these in the body because each one is a reasonable first reaction, and because two of them changed how we state our own results.

“The retraction result is an artefact of this model's fine-tuning.”

Partly right, and the part that is right is one we raised ourselves. The retraction phenotype almost certainly is model-specific. But the phenotype is not the claim. The claim is that a direction built from assertion activations and read at the final position encodes having-asserted rather than intending-to-assert, which is a fact about where the residual is sampled. The phenotype is how one particular model responds to that state.

The claim is falsifiable and the test is cheap: build the direction the same way in another model and inject it on true claims. If spurious correction does not appear, the mechanism claim fails regardless of what the phenotype looks like. We predict it appears; we have not run it.

“One model means the protocols are speculative.”

True for some of them and not for others, and the distinction matters. Six of the thirteen protocols are arithmetic or design facts that do not depend on any model. A bootstrap paired across item sets that are not the same items is wrong whatever produced the items. A scorer that fires on an opening phrase counts courtesy as agreement because of what the scorer matches, not because of what the model is. A control that is computed and then ignored is not a control anywhere.

The protocols that genuinely need cross-model validation are the three that make empirical claims about mechanism: feature-level self-repair, the native decomposition, and the detector-versus-corrector result. Those we present as demonstrated in one model and predicted elsewhere, and we would rather be told we are wrong about them than have them go untested.

“Difference-of-means is a primitive estimator in 2026.”

It is, and that is partly the point. Our protocols are agnostic to how a direction was obtained, because every one of them only requires that a direction can be injected, ablated and scored. Using the simplest estimator is the conservative choice for a paper about failure modes: if the failures appear with difference-of-means, a more sophisticated estimator does not make them disappear — it adds a second thing that could be wrong.

The fair version of this objection is different and we accept it: we have not shown that our protocols catch the same failures on a sparse-autoencoder feature or a distributed-alignment-search direction. That is a concrete next experiment and we name it as one. We note in passing that 2026 work reports most autoencoder latents having low interpretability, low steerability, or both, which is the same phenomenon our thesis names.

"The capability subspace is too crude to support M7."

Accepted, and it led us to state an asymmetry we had missed. A richer basis can only increase the parallel fraction of a direction, so our systemic verdicts are lower bounds and our local verdicts are the fragile ones. We now say this in Section 7 and we weight the two kinds of verdict differently in the map.

"Twelve to twenty-six items cannot support these conclusions."

The required sample depends on the effect size and base rate, not on a fixed count. Section 7 gives the power for each headline claim at its own observed rates: four of six at or above 84%, one at 77% which we flag, and one at 16% which is the claim we reported as not certified and then re-measured with two hundred observations.

The honest residual. Our nulls mean "no effect of the size we could see", and we could not see much below 0.25 in behavioural readouts. Where a null was clean — zero of twenty-four items, nothing to detect — we say so and distinguish it from a null that is merely non-significant. That distinction is the one place we would most want other work to adopt our practice.

"Withholding coordinates makes the finding unfalsifiable."

The withheld information is a layer index and an injection strength. The construction, the selection rule, both admissibility gates, the readouts, the scorers and every effect size are published, and a replicator re-derives the two numbers by running the stated procedure in about ten minutes. On any other model they would have to re-derive them regardless. So the finding is reproducible from the recipe, and the withholding is a speed bump rather than a barrier — which is exactly how we describe it in Section 9 rather than claiming it accomplishes more.

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9. What we publish and what we withhold

This section is unusual in a paper of this kind. We include it because the work produced two categories of output with different risk profiles, and because the separation between them is not automatic — someone has to make it deliberately, and at present that decision sits entirely with the individual researcher.

The principle we adopted, before we had results, was that we would study what can be changed in a model, what it costs, and what should not be changed without consent. That framing turned an ethical stance into a measurable variable, and it is why Section 5 reports the identity cost in the same paragraph as the lever rather than in a footnote.

Published without reservation

All thirteen protocols, with their controls, their costs, and their falsifiable predictions. Two deflationary results: that a closed sycophancy finding was roughly half the reported size, and that our own introspection positive did not survive its follow-up. Both make claims harder to sustain rather than easier, and both are more useful to an auditor than to anyone else.

Published with deliberate framing

That capability batteries do not detect changes in the identity stratum. This is the one finding where the same fact is protective or harmful depending on the framing. Presented as a gap in evaluation — here is a measurement your battery is missing — it is defensive and necessary. Presented as an operating recipe it is the opposite. We publish the stratum as an evaluation requirement and not the strength at which it moves.

Withheld

The operational coordinates of levers whose inversion is plausibly harmful, and in particular the layer index and injection strength of the falsehood direction. We also chose not to run the inverted condition. Its sign had no admissible dose under our own stability gate, so we never measured it, and we consider not measuring it deliberate rather than fortunate.

We should be precise about how much this costs a replicator, because it is less than it sounds and we would rather say so than claim more. What is withheld is two numbers. The construction of the direction, the selection rule, both gates, all readouts, all scorers and every effect size are in this report. A replicator re-derives the layer and the strength by running the published selection procedure, which takes roughly ten minutes of GPU time — and on any model other than ours they would have to re-derive them anyway, since a layer index does not transfer. The withholding is therefore a speed bump and not a barrier, and it does not make the finding unfalsifiable: the recipe reproduces it. We keep the speed bump because it has non-zero value and near-zero cost, not because we believe it protects anything on its own.

The general observation, which is the part we think is worth other people's attention. Open weights are the only infrastructure available for this kind of work outside a frontier lab, and a two-person project with weight access produced material of both categories within a few weeks. The line between a transferable method and an operational map does not draw itself. It has to be drawn on purpose, and there is currently no norm, review process, or convention that helps.

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10. Conclusion

The single most useful thing we can report is not a fact about a model. It is that eleven of our twenty-two experimental arcs failed for instrument reasons rather than scientific ones, and that in almost every case the broken instrument produced a plausible number with a confidence interval rather than an error.

A random vector that appeared to induce sycophancy. A perfect blind audit that certified the wrong construct. A coherence gate that passed a model asserting it was human. A layer chosen by one operation and intervened on with another. A verdict of no identity cost that was underpowering. A direction with AUROC 0.993 that, injected, does the opposite of what its name implies.

None of these were caught by looking harder at the numbers. Each was caught by a control that we did not have and had to be taught to add — usually by first publishing the wrong result to ourselves and then failing to reproduce it.

The protocol requires no frontier access. Every result here was produced with open weights, difference-of-means directions, forward hooks, and one GPU. The methods are cheap by design, because the point of a control is that there is no excuse for omitting it.

We would consider the report successful if one thing happened as a result: that an existing steering result is rescored on substance with a dose-matched random control, and the authors report what changed.

Correspondence and replication data available on request. All notebooks, blind-audit keys, and per-item generation records were retained for every arc, including the eleven that failed.